

Statement of Purpose

My name is **Aaditya Sattawan**, and I am pursuing an Integrated Dual Degree (B.Tech + M.Tech) in Mechanical Engineering at IIT (BHU) Varanasi. While my academic background is not traditionally associated with Artificial Intelligence and Machine Learning, I have always been fascinated by solving challenging problems and learning new technologies.

My journey into Machine Learning began during “**Enigma Codefest'26**”, a machine learning hackathon organized at IIT (BHU). At the time, I had little prior exposure to ML and participated purely out of curiosity. Over three days, I immersed myself in understanding data, experimenting with models, and exploring concepts far beyond my coursework. My team secured **2nd place**, but more importantly, I discovered a field that combined analytical thinking, creativity, and real-world impact in a way I had never experienced before.

Since then, I have actively pursued Machine Learning through competitions and projects. I secured a **Top 10 finish in “MLX Session Zero”**, where I worked on solar radiation prediction and gained hands-on experience with feature engineering, model selection, hyperparameter tuning, and evaluation techniques. These experiences taught me that successful ML solutions are not just about achieving high scores but about solving meaningful real-world problems.

This mindset led me to build **InterviewIQ**, an AI-powered interview preparation platform inspired by a challenge many non-circuitual students face, understanding how well their profile aligns with industry expectations. The platform analyzes resumes against job descriptions, identifies skill gaps, generates personalized interview questions, and creates AI-powered preparation roadmaps. Building InterviewIQ showed me how AI can be used to create practical solutions that directly help people.

Currently, I am developing an **AI-powered Candidate Ranking Engine** for the “**India.Runs**” Hiring Challenge, organized by **Redrob**. The system combines semantic search, transformer-based retrieval, reranking techniques, and explainable AI to intelligently rank candidates against job descriptions. Through this project, I have progressed from applying machine learning models to designing complete AI systems that address practical recruitment challenges at scale.

Beyond AI, I enjoy solving complex problems and have achieved a **Codeforces Specialist rating of 1501**. Competitive programming has strengthened my problem-solving and analytical thinking skills. As **Captain of the IIT (BHU) Aquatics Team and an Inter-IIT medalist**, I have also developed leadership, teamwork, discipline, and the ability to perform under pressure.

While my projects have provided practical exposure, I want to deepen my understanding of the theoretical foundations of Machine Learning, particularly optimization, representation learning, deep learning architectures, and model generalization. Amazon ML Summer School offers an exceptional opportunity to learn these concepts from leading researchers and practitioners while interacting with a community of passionate learners.

I believe my combination of strong problem-solving skills, hands-on AI experience, leadership, and relentless curiosity makes me a strong fit for the program. I am eager to learn from the best, contribute meaningfully, and use that knowledge to build impactful AI solutions that solve real-world problems.